

Metals / Non-metals:

thermal conductivity → only metals

electrical conductivity → only metals

malleability / ductility → only metals

melting / boiling points → metals have higher MP / BP

Metal reactions

• Metals + Acid → Salt + Hydrogen

Sodium + ^{hydrochloric} acid → ^{sodium} chloride + hydrogen

• Metals + Water → Metal Hydroxide + Hydrogen

sodium + water → sodium hydroxide + hydrogen

• Metals + Oxygen → Metal oxide

copper + oxygen → copper oxide

Uses of metals

Aluminium

* manufacturing of aircrafts [low density]

* manufacturing of overhead cables [low density / good conductor]

* in food containers [resistance to corrosion]

Copper

* electrical wiring [good conductivity / ductility]

Alloys

• a mixture of a metal with other elements

brass → mixture of copper & zinc

stainless steel → mixture of iron and chromium / nickel / carbon etc.

* alloys can be harder and stronger than pure metals and are more useful

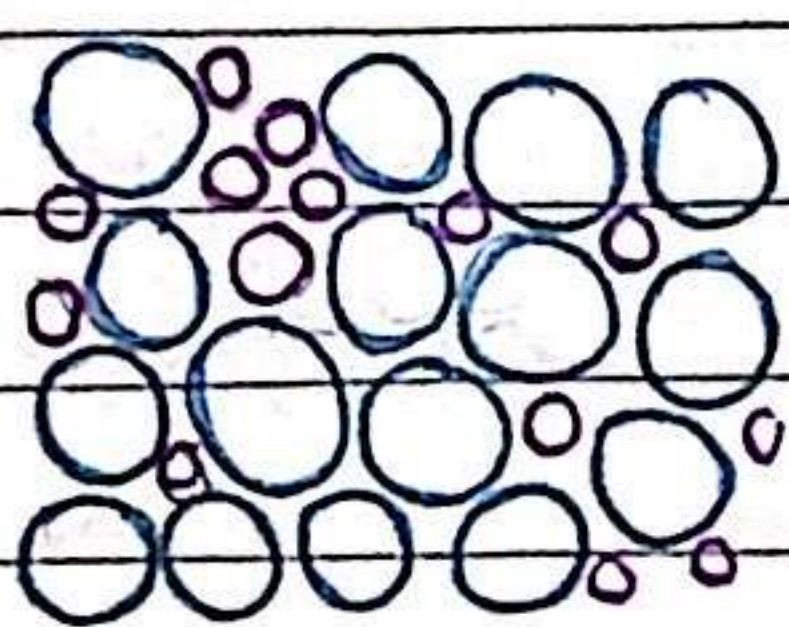
* alloys are harder/stronger because they have different sized atoms which means the layers can no longer slide over each other.

Uses of alloys

stainless steel

* in cutlery [hard & resistant to rusting]

Diagram



if force is applied, it cannot make the alloy particles slip easily.

* metal atoms form positive ions by loss of electrons when they react with other substances.

* the tendency of a metal is how reactive the metal is ^{to lose electrons}

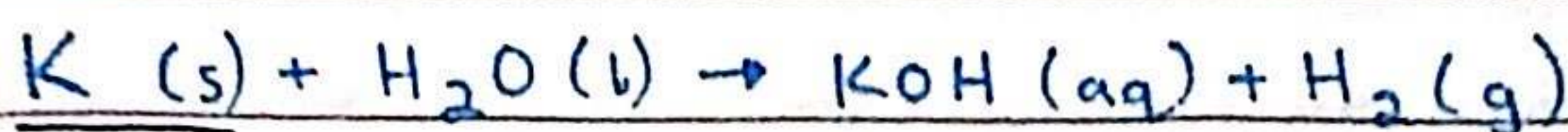
* a metal high up on the series loses electrons easily thus is more reactive.

→ any metal displaces another metal that is below it in the reactivity series.

Magnesium + iron (II) sulfate → magnesium sulfate + iron
 replaces iron

Zinc + Magnesium chloride → No reaction

magnesium is more reactive than zinc so no reaction takes place



more reactive
than hydrogen

replaced

} water



more reactive
than hydrogen

replaced

} steam



more reactive
than hydrogen

replaced

} acid

Deducing order of reactivity

- can be deduced by making experimental observations
- more vigorous the reaction, higher up the reactivity series the metal is

Reactivity of aluminium :

- high in reactivity series but doesn't react with water and slow reactions with dilute acids
- because it readily reacts with oxygen, forming a protective layer of Aluminum oxide (thin)
- this layer prevents reaction with water and acids

Rusting

- oxygen + water must be present for rusting
- reaction between iron, water and oxygen

Barrier method

- coating the iron with paint, grease, plastic or zinc prevents it from rusting
- this prevents oxygen/water to touch the iron
- if any other metal rather than iron & steel, oxidise in the air, and break down, that metal will have corroded

* a more reactive metal can be attached to a less reactive one so the more reactive corrodes, protecting the less reactive metal.

* zinc is more reactive than iron/steel thus it can be used to prevent rusting.

Galvanising: layer of zinc put on iron/steel

→ done by electroplating or dipping it in molten zinc

→ $ZnCO_3$ is a reaction of zinc with the air which protects the iron

→ this is called sacrificial protection

Extraction of metals

- Earth's crust contains metals and metal compounds
- Useful metals are often chemically combined with other substances forming ores which is a rock that contains enough metal to make it worth extracting.
- extracted by electrolysis, blast furnace or reacting with a more reactive metal
- more reactive metals are extracted through electrolysis while less reactive are done so by a blast furnace

Extraction of iron (blast furnace)

raw materials:- hematite, carbon (coke), limestone, hot air

① Carbon + Oxygen $\xrightarrow{\text{heat}}$ Carbon dioxide

② Carbon dioxide + Carbon $\rightarrow$ Carbon monoxide strong reducing agent!!

③ Carbon monoxide + Iron oxide $\rightarrow$ Iron + Carbon dioxide

① $C + O_2 \xrightarrow{\text{heat}} CO_2$

② $C + CO_2 \rightarrow 2CO$

③ $3CO + Fe_2O_3 \rightarrow 2Fe + 3CO_2$

① Limestone $\rightarrow$ Calcium oxide + Carbon dioxide

② Calcium oxide Silicon oxide $\rightarrow$ SLAG

① $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$ $\leftarrow$ thermal decomposition

② $\text{CaO} + \text{SiO}_2 \rightarrow \text{CaSiO}_3$ $\leftarrow$ neutralization reaction

Extraction of aluminium (electrolysis)

- main ore of aluminium is known as bauxite

- * First the bauxite is purified to produce Al_2O_3

- * then the Al_2O_3 is dissolved in molten cryolite

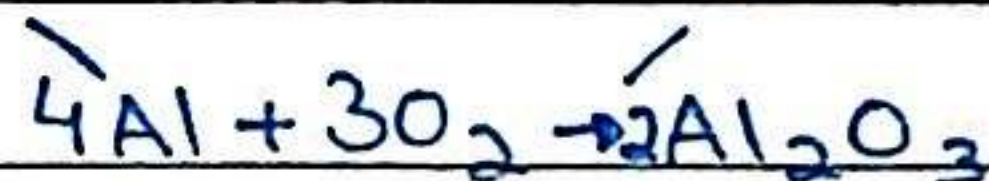
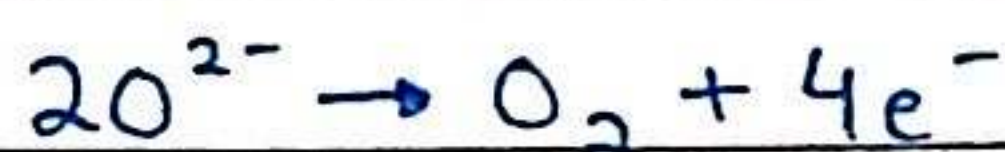
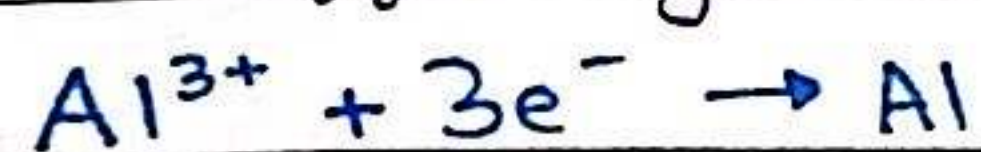
- $\rightarrow$ which reduces the melting point

- * the mixture is put in an electrolysis cell

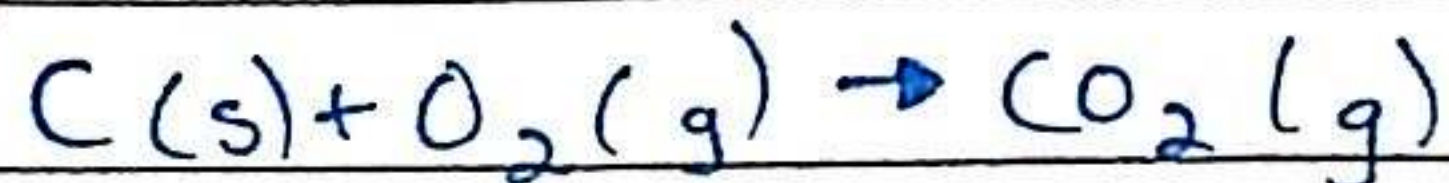
- $\rightarrow$ made from steel, lined with graphite

- * at the cathode, aluminium ions gain electrons and molten aluminium forms at the bottom of the cell

- * at the anode, oxide ions lose electrons and oxygen gas is produced



- * The carbon (C) in the graphite anode reacts with the oxygen produced by the oxide to produce CO_2



$\rightarrow$ As a result, the anode wears away and has to be replaced regularly

(A lot of electricity is required for this process of extraction)